

NICOLAS RUTH

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EDUCATION

University of Denver — Denver, CO Expected May 2027

B.S. in Electrical Engineering

Relevant Coursework: Advanced Electronics, Physical Electronics, Signals & Systems, Microprocessors, Electromagnetics, Embedded Systems

TECHNICAL PROJECTS

Electromagnetic Rail Launcher (Railgun) Power Electronics · High Voltage

- Built a rail-type electromagnetic launcher driven by a DC boost converter stepping an 11.1V LiPo up to ~225V to charge a 13,200 μF capacitor bank (six 250V / 2200 μF electrolytics in parallel, ~330 J stored), discharged across dual aluminum rails to accelerate a small metal projectile.
- Designed a 2.6 k Ω inrush-limiting resistor network that utilized a 600V blocking diode to protect the charging stage.

Electromagnetic Pulse (EMP) Demonstrator High Voltage

- Designed a capacitor-discharge EMP generator using a 9V high-voltage step-up converter to charge a 20 kV ceramic capacitor bank, discharged through a hand-wound coil to produce a pulsed magnetic field.

FM Radio Receiver RF / Analog

- Built an FM radio receiver from simple components: a hand-wound 0.15 μH coil, a 1–30 pF variable capacitor that tunes the 88–108 MHz band, an S9018 RF transistor recovers the signal, and an LM386 amplifies the audio to a speaker.

PCB DESIGN

DemoSat High-Altitude Balloon Payload KiCad

- Designed and soldered a sensor-payload PCB (Arduino Uno shield) in KiCad, integrating temperature (DS18B20), humidity (HIH9120), barometric-pressure (DPS310), and 3-axis accelerometer (ADXL345) sensors over I²C and SD logging; selected components to meet the flight's altitude, temperature, and low-pressure requirements.

Battery Tester Board KiCad, LPKF CNC

- Designed a custom battery-tester PCB in KiCad, creating non-standard component footprints and drill specifications for in-house fabrication on an LPKF CNC milling machine.

Arduino Uno I/O Shield KiCad

- Built an Arduino Uno I/O shield integrating potentiometers, push buttons, status and RGB LEDs, an OLED display, and selectable-voltage I²C / SPI expansion headers.

TECHNICAL SKILLS

Languages: C / Embedded C, Python, MATLAB, Java

Tools & Software: KiCad, LTspice, STM32CubeIDE/ArduinoIDE, SolidWorks/Onshape, Microsoft Office

Hardware & Lab: STM32, Arduino, and Teensy microcontrollers, PCB schematic and layout design, component selection, soldering, oscilloscope, multimeter, 3D printers, hand & power tools

Concepts: Analog & digital circuit design, PCB design & fabrication, sensor integration & data logging, power electronics, electromagnetics, embedded systems

LEADERSHIP & ACTIVITIES

Member — DU IEEE Chapter 2026 - Present

- Assisted in designing a PCB layout for a slice of the LED matrix cube.

DU Time Attack Motorsport Team 2025

- Attended the team's aerodynamics meetings, following vehicle aero and downforce design discussions.

Mechanical Team Lead — FIRST Robotics 2021 – 2023

- Led the mechanical subteam, training new members on machines, components, and tools to deliver a competition-ready robot.

Member — Mile High Flight Club 2022 – 2023

- Studied aviation, flight principles, and aerospace engineering careers through regular meetings.

Member — Colorado Young Leaders 2023

- Organized community service events focused on environmental sustainability.

WORK EXPERIENCE

Lead Cashier — Miss Molly's Bakery, Denver, CO 2022 – 2023

- Sole closing employee; trained new cashiers and streamlined checkout to cut customer wait times.

Landscaper — Self-Employed / Local Clients, Denver, CO 2022 – 2023

- Maintained grounds and safely operated landscaping equipment while managing client schedules independently.